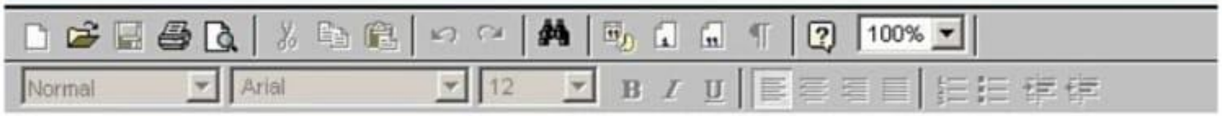
[redacted]

Question No : 50 of 52

What is virtual function? Explain it with coding example.

Answer (Please click here to Add Answer)



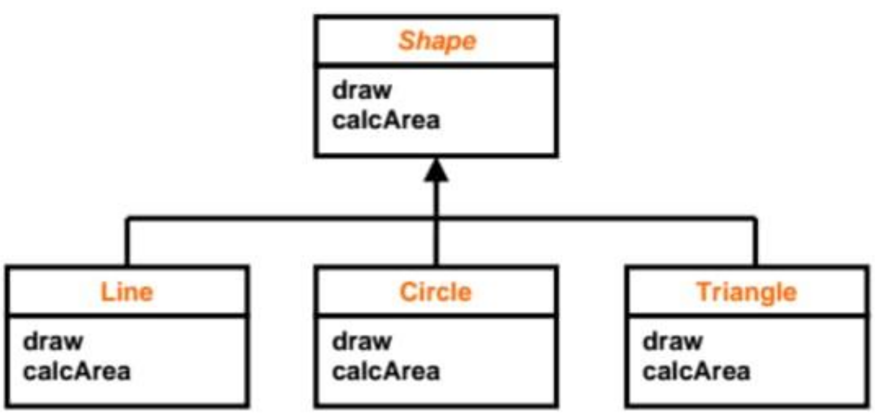
28.2.Virtual Functions:

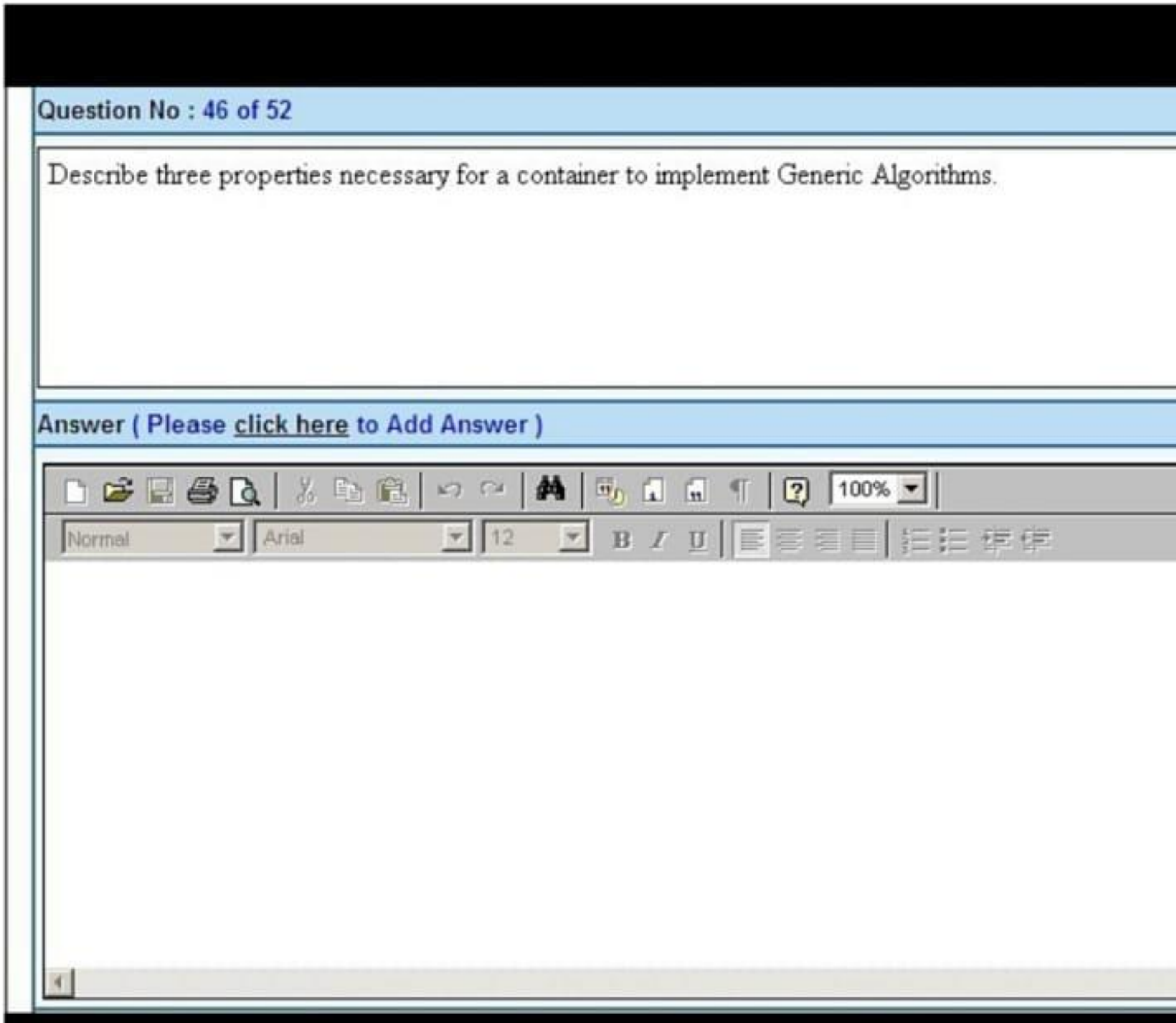
Virtual functions achieve exactly same kind of functionality that was achieved in above code with complex code of switch statement.

- Target class of a virtual function call is determined at run-time **automatically**.
- In C++, we declare a function virtual by preceding the function header with keyword "virtual"

Now we see how we can use virtual functions in case of our shape hierarchy,

28.3.Shape Hierarchy






```
{
for (int i = 0; i < strlen( str1 ) && i < strlen( str2 ); ++i)
if ( str1[i] != str2[i] )
return str1[i] - str2[i];
return strlen(str1) - strlen(str2);
}
```

6. Describe three properties necessary a container to implement generic algorit

Answer:- (Page 301)

We claimed that this algorithm is generic, because it works for any aggregate object (container) that defines following three operations

a. Increment operator (++)

b. Dereferencing operator (*)

[redacted]

Question No : 47 of 52

Mobile is a good entity for being an object. List down any two characteristics, behavior and one unique ID in this context

Answer (Please click here to Add Answer)

Normal Arial 12 B I U 100%

[redacted]

7. Mobile is a good entity for being an object. Write characteristics, behavior and unique ID.

Answer:-

Characteristics: Buttons, Battery, Screen

Behavior: Calling, Sending Message, Internet, Multimedia

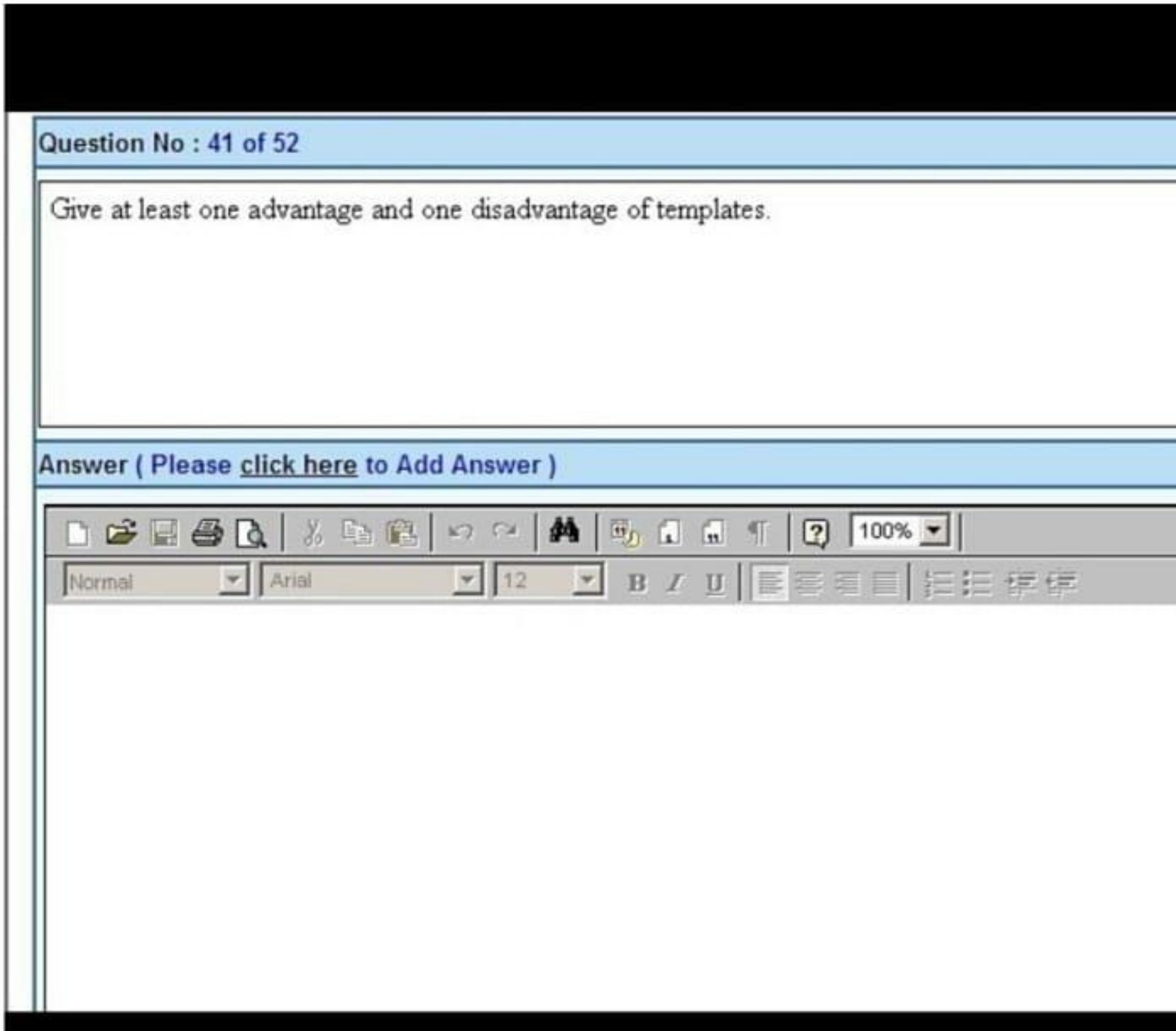
Unique ID: Nokia

8. What are the container requirements? Write details.

Answer:-

As each container need to perform certain operations on the elements added to it like their copy while creating another instance of container or their comparison while performaing certain operation on them like their sorting , so the elements that we are going to add in containers should provide this kind of basic functionality. Examples of these functionalities are given below,

Muhammad Moaaz Siddiq – MCS (2nd)
mc100401285@Gmail.com
Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari



The least one advantage and Disadvantage of Template
Answer:- (Page 300)
Advantages:
Templates provide
• Reusability
• Writability

Disadvantages:
• Can consume memory if used without care.

FINALTERM EXAMINATION 2011

1-define composition and give its example with coding
Answer:- (Page 53)
An object may be composed of other smaller objects, the relationship between the "part" objects and the "whole" object is known as Composition.

Question No : 44 of 52

Give two advantages that Iterators provide over Cursors.

Answer (Please click here to Add Answer)

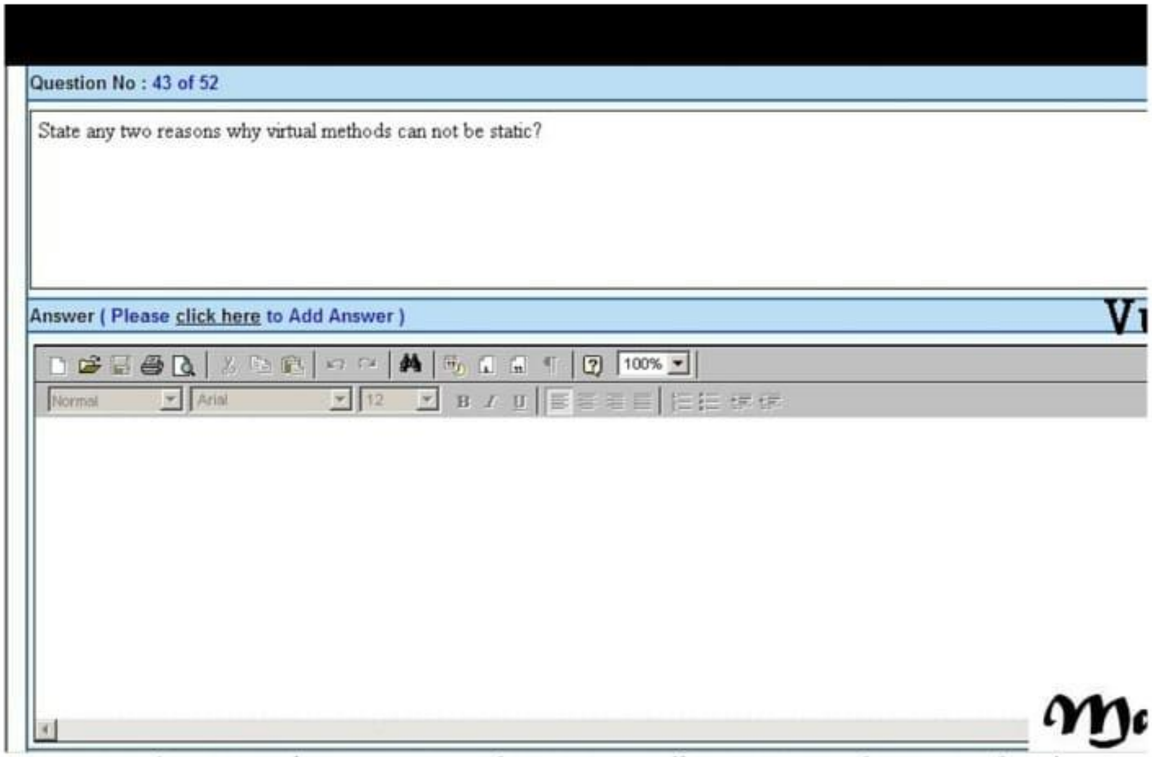
Q5) Give three advantages that Iterators provide over Cursors.

Answer: - (Page 311)

a. With Iterators more than one traversal can be pending on a single container
b. Iterators allow to change the traversal strategy without changing the aggregate object
c. They contribute towards data abstraction by emulating pointers

Q6) Consider the code given below explain what kind of association exists between class A and class B Justify your answer as well.

```
class A{

private:
int a,b,c;

public:
....
};

class B{

private:
int d,e,f;
A obj1;

public:
```



Q4) State any two reasons why the virtual methods can not be static?

Answer:- **Click here for detail**

The virtual method implies membership, so a virtual function cannot be a nonmember function. Nor can a virtual function be a static member, since a virtual function call relies on a specific object for determining which function to invoke. A virtual function declared in one class can be declared a friend in another class.

Q5) Give three advantages that Iterators provide over Cursors.

Answer: - (Page 311)

a. With Iterators more than one traversal can be pending on a single container
b. Iterators allow to change the traversal strategy without changing the aggregate object
c. They contribute towards data abstraction by emulating pointers

Q6) Consider the code given below explain what kind of association exists between class A and class B. Justify your answer as well.

```
class A{

private:
int a,b,c;

public:
....
};

class B{

private:
int d,e,f;
A obj1;
```

uestion No : 42 of 52

Why we need to use friend functions when overloading insertion and extraction operators?

nswer (Please click here to Add Answer)

cin and **cout** are objects of istream and ostream classes used for input and output respectively, the insertion and extractions operators have been overloaded in istream and ostream classes to do these tasks.

When we write lines like,
int i;
cin>> i;
cout << i;

Actually we are using istream and ostream class objects and using these objects we are calling these classes overloaded (>> and <<) operators that have been overloaded for all basic types like integer, float , long , double and char *.

[redacted]

Question No : 41 of 52

Sort the following data in the order in which compiler searches a function?
Complete Specialization, Generic Template, Partial Specialization, Ordinary Function.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

9. Sort data in the order in which compiler searches a function. Complete specialization, generic template, Partial specialization, Ordinary function. (2Marks)

Answer:- (Page 286)

a. First of all compiler looks for complete specialization

b. If it can not find any required complete specialization then it searches for some partial specialization

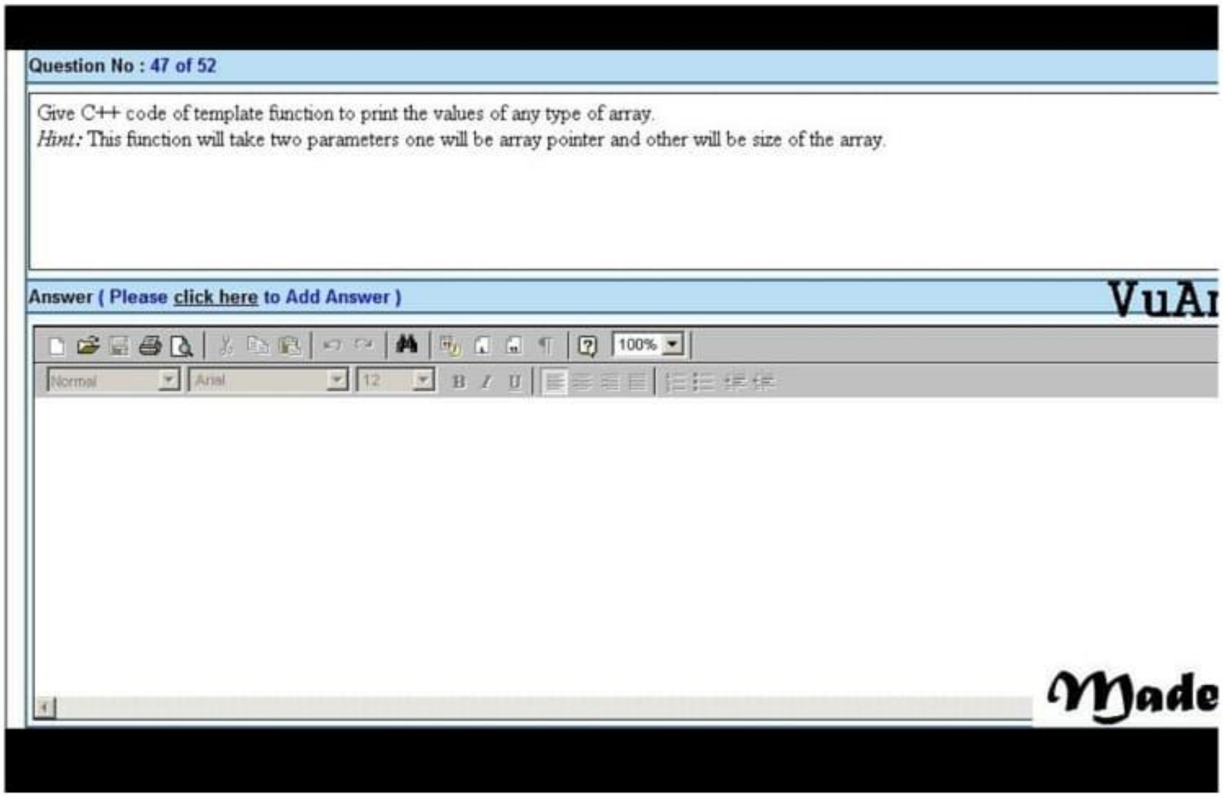
c. In the end it searches for some general template

10. Give the names of two types of containers basically known as first class containers.(2 Marks)

Answer:- (Page 317)

Sequence and associative containers are collectively referred to as the first-class containers

Muhammad Moaaz Siddiq – MCS (2nd)
mc100401285@Gmail.com
Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari



Give the C++ code of template function to print the values of any type of array I int.this function will take 2 parameters one will be pointer and other will be will be size of array (mrk3)

Answer:- (Page 257)

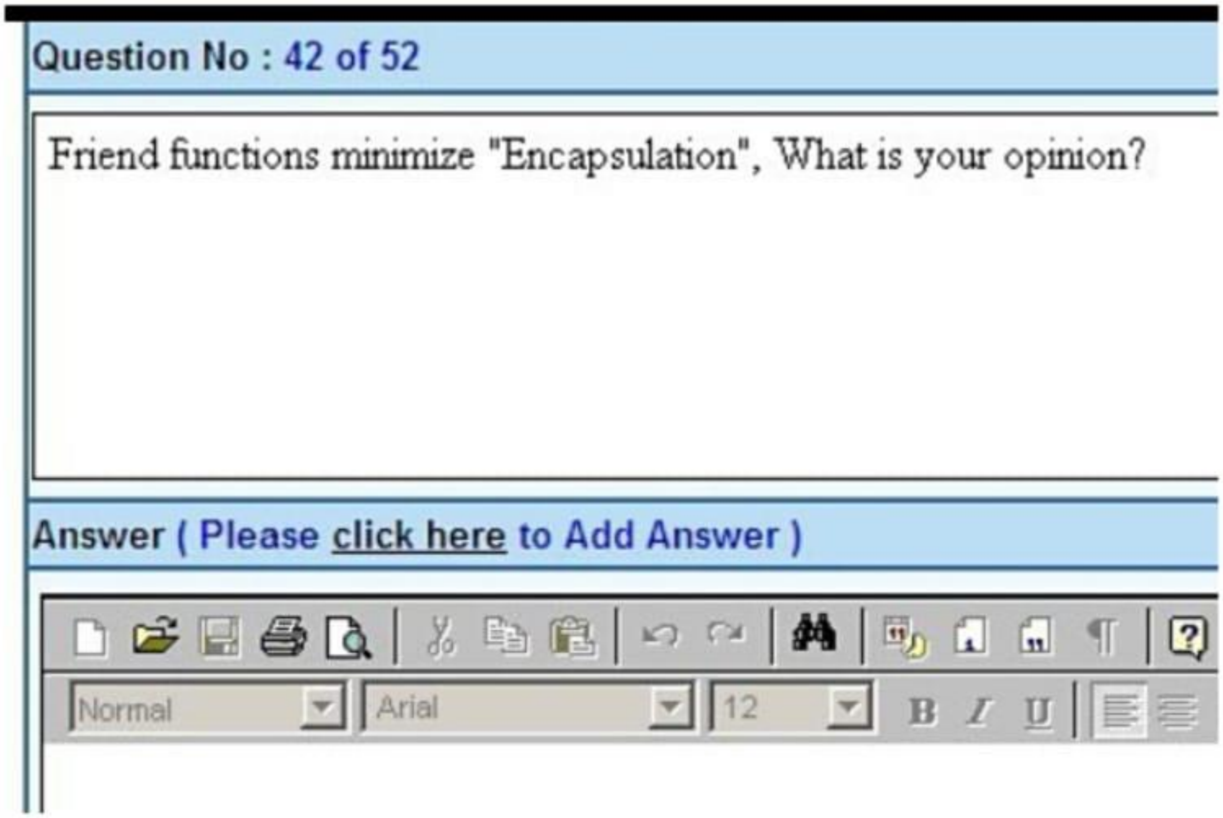
```
template< typename T >
void printArray( T* array, int size )
{
for ( int i = 0; i < size; i++ )
cout << array[ i ] << ", "; // here data type of array is T
}
```

Give the name of three operation that a corsor or itrator generally provide (mrk3)

Answer:- (Page 305, 309)

- T* first()
- T* beyond()
- T* next(T*)

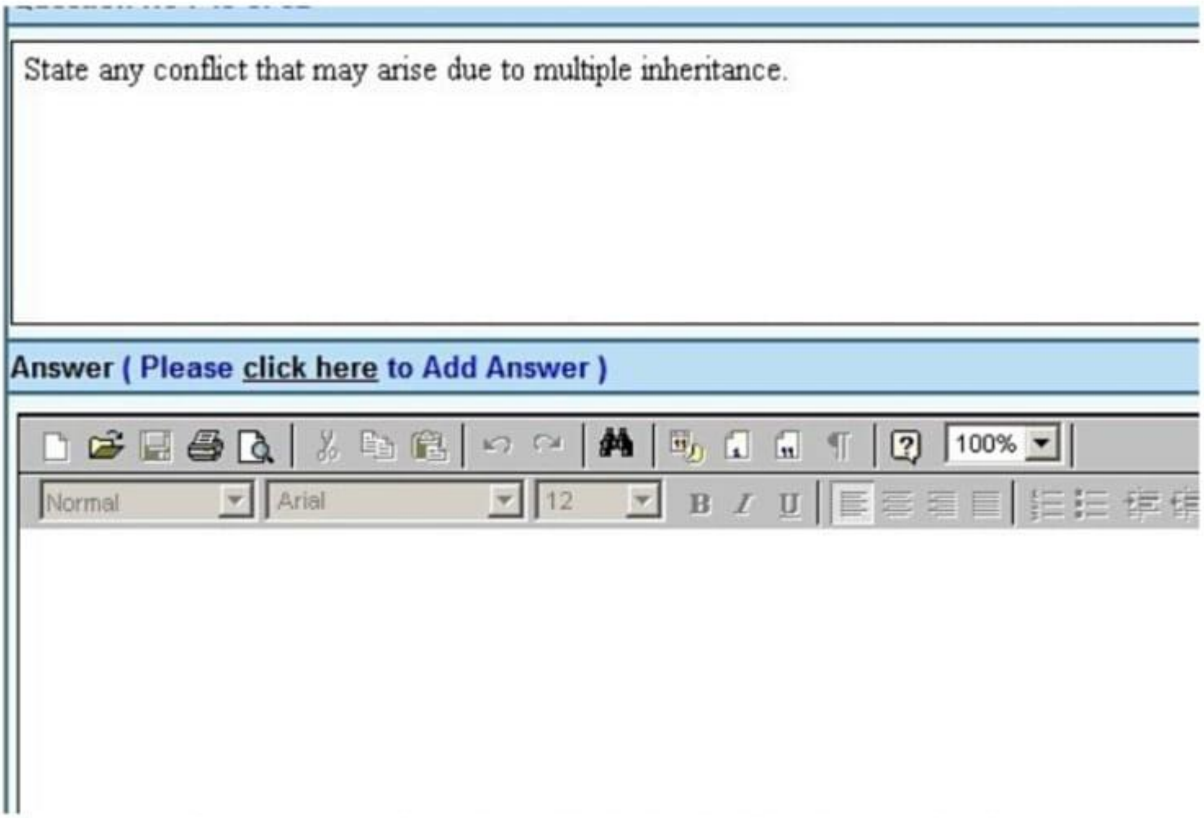
Muhammad Moaaz Siddiq – MCS (2nd)
mc100401285@Gmail.com
Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari



18.3. Friend Functions and Operator overloading

Friend functions minimize encapsulation as we can access private data of any class using friend functions,
This can result in:

- Data vulnerability
- Programming bugs
- Tough debugging



Give the name of two basic types of containers collectively called First class containers?
Answer:- rep

Muhammad Moaaz Siddiq – MCS (2nd)
mc100401285@Gmail.com
Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari

Question No: 30 (Marks: 2)
State any conflict that may rise due to multiple inheritance?
Answer :-(page 248)
If more than one base class has a function with same signature then the child will have two copies of that function.Calling such function will result in ambiguity.

Question No: 31 (Marks: 3)
What will be the output after executing the following code?

[redacted]

Question No : 41 of 52

What are the non - type parameters for templates?

Answer (Please click here to Add Answer)

Normal Arial 12 100%

[redacted]

```
catch (…) {
}
```

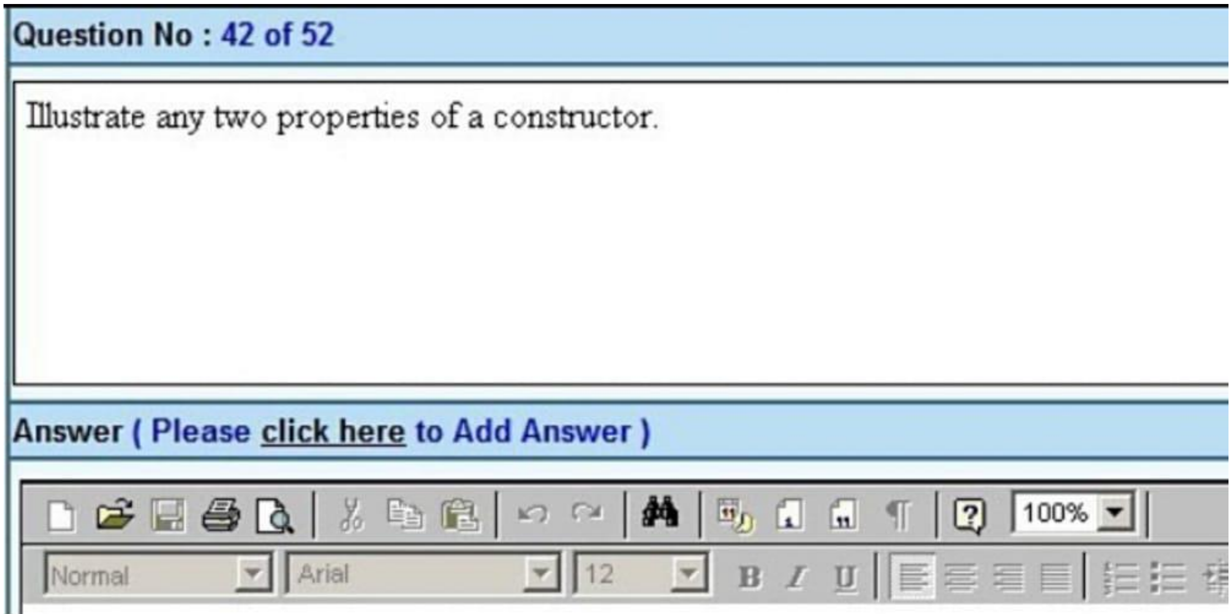
FINALTERM EXAMINATION 2012 (FEB)

1. How can we set the default values for non type parameters?

Answer:- (Page 284)

We can set default value for this non type parameters, as we do for parameters passed in ordinary functions,

```
template< class T, int SIZE = 10 >
class Array {
private:
T ptr[SIZE];
public:
void doSomething();
…
}
```


Question No : 42 of 52

Illustrate any two properties of a constructor.

Answer (Please click here to Add Answer)

7-what is constructor?

Answer:- (Page 74)

Constructor is used to **initialize** the objects of a class. Constructor is used to ensure that object is in well defined state at the time of creation.

8-define static and dynamic binding.

Answer:- (Page 227)

Static binding means that target function for a call is selected at compile time

Dynamic binding means that target function for a call is selected at run time

FINALTERM EXAMINATION 2011

Question No : 52 of 52

```
void sample_function(double test) throw (int)
{
        cout << "Starting sample_function.\n";
        if(test < 100)
          throw 42;
}
```

Answer (Please click here to Add Answer)

Question No: 35 (Marks: 5)

What is the output produced by the following program?

```
#include<iostream.h>

void sample_function(double test) throw (int);

int main()
{
        try
        {
                cout <<"Trying.\n";
                sample_function(98.6);
                cout << "Trying after call.\n";
```

Muhammad Moaaz Siddiq – MCS (2nd)
mc100401285@Gmail.com
Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari

```
        }
        catch(int)
        {
                cout << "Catching.\n";
        }

        cout << "End program.\n";
        return 0;
}
void sample_function(double test) throw (int)
{
        cout << "Starting sample_function.\n";
        if(test < 100)
          throw 42;
}
```

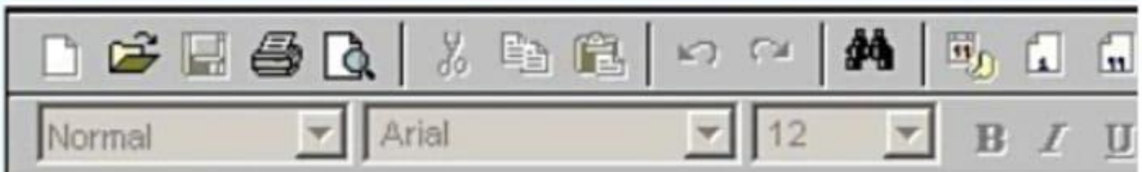
Answer:-

Trying.
Starting sample_function.
Catching.
End program.

Question No : 48 of 52

Give the general syntax of nested try catch blocks?

Answer (Please click here to Add Answer)



Stack Unwinding can take place in the following two ways,

1. When we have nested try catch blocks (one try catch block into other try catch block), for example

```
try {
                    try {

                    } catch( Exception e)  {

                  }
```

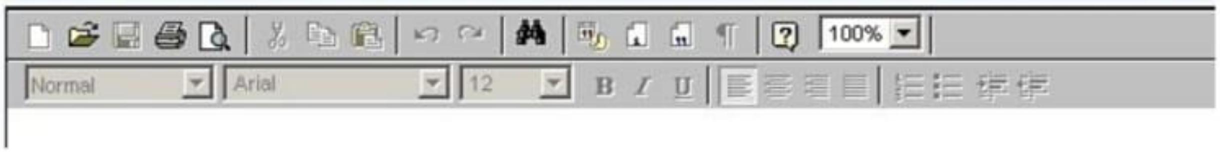


```
 } catch(exception e){

}
```

Question No : 47 of 52

In which situation do we need to implement virtual inheritance? Explain with the help of an example.

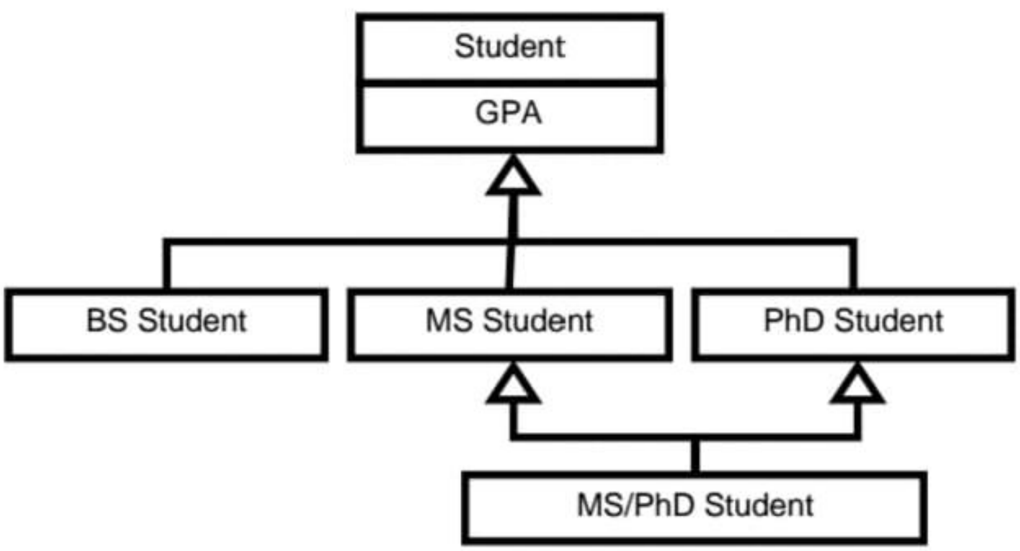
Answer (Please click here to Add Answer)

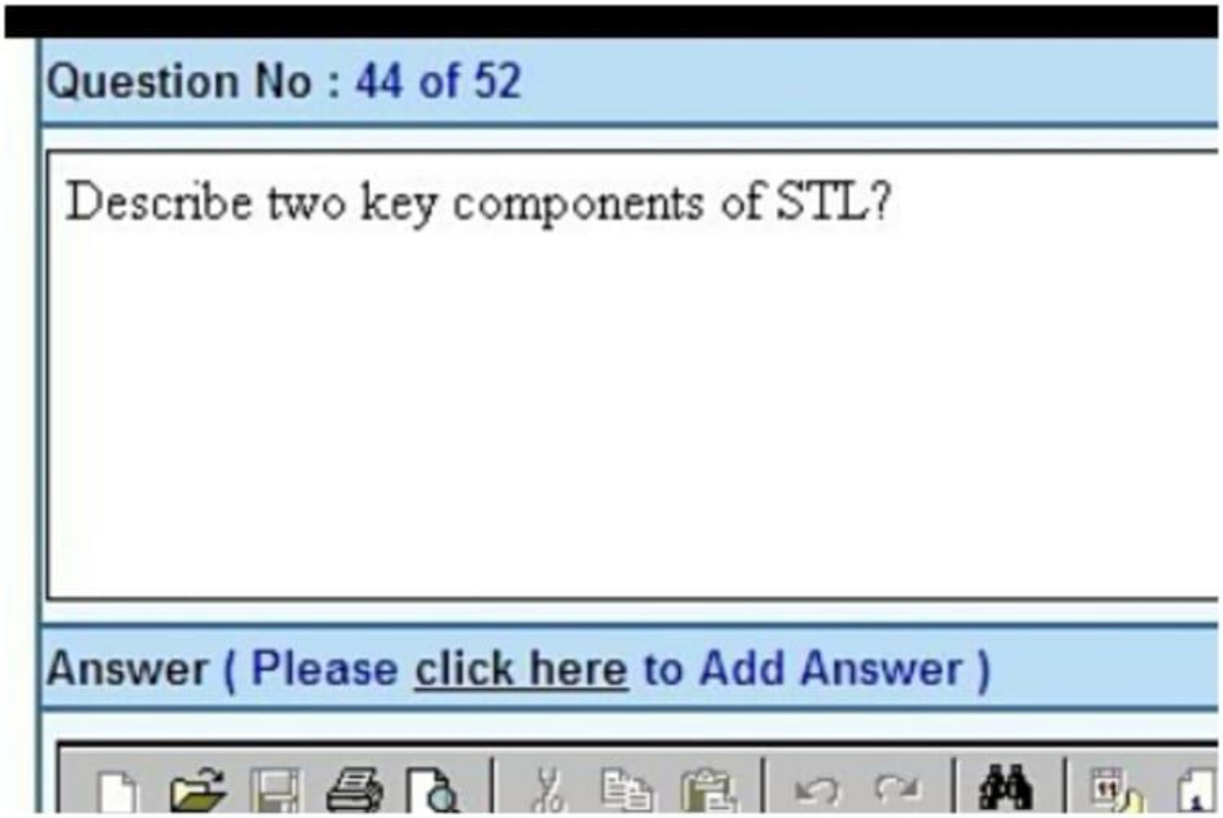


When to use Virtual Inheritance?

Virtual inheritance must be used when necessary. It can be used in the situations when programmer wants to use two distinct data members inherited from base class rather than one.

Example




Question No : 44 of 52

Describe two key components of STL?

Answer (Please click here to Add Answer)

Standard Template Library

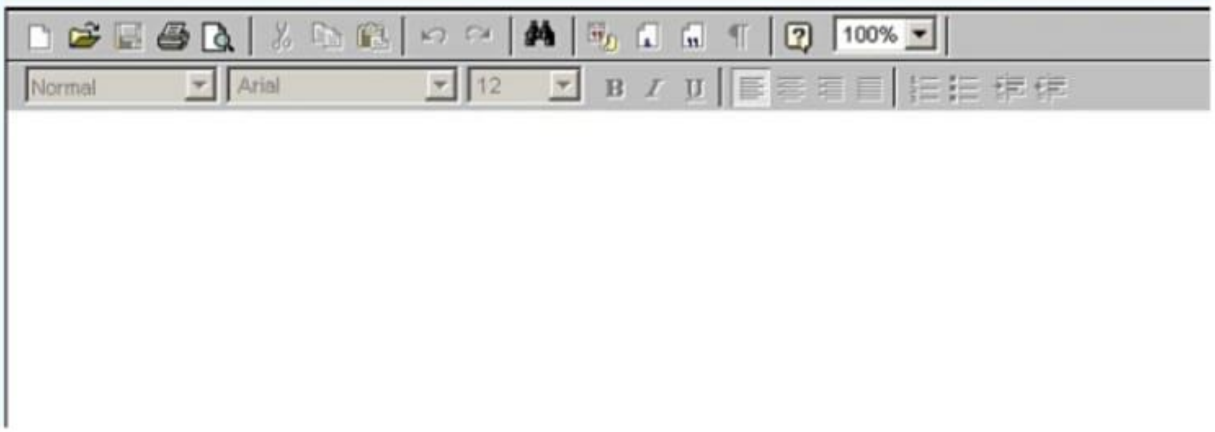
STL consists of three key components

- Containers
- Iterators
- Algorithms

Question No : 43 of 52

How static and dynamic binding is used in the case of virtual functions? Justify your answer as well.

Answer (Please click here to Add Answer)

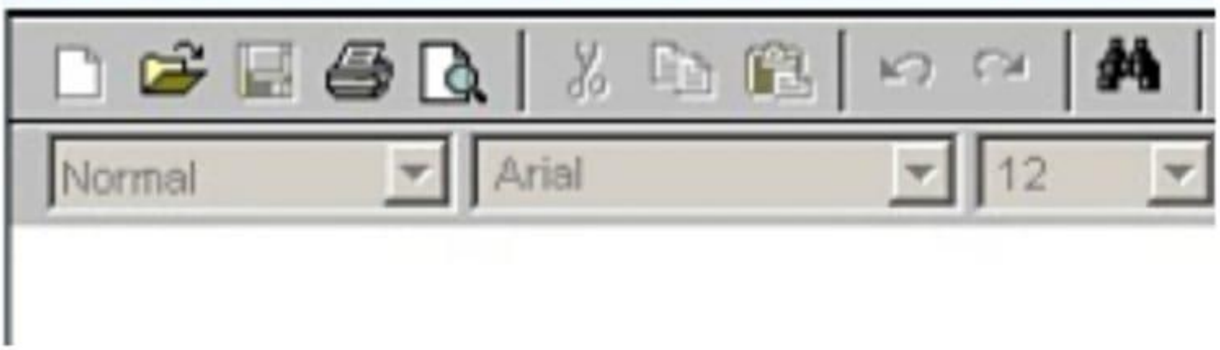


28.4.Static vs Dynamic Binding

- **Static** binding means that target function for a call is selected at compile time
- **Dynamic** binding means that target function for a call is selected at run time

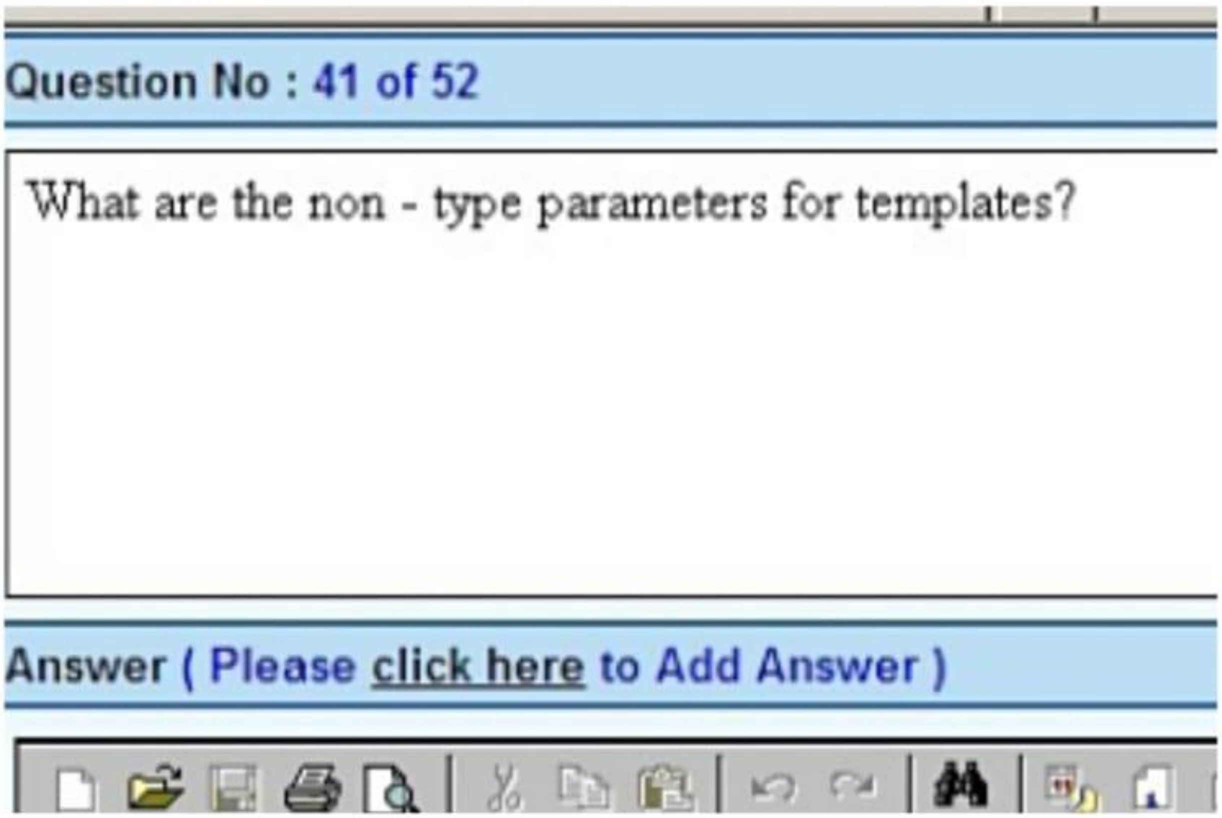
Define Catch Handler in berief.

Answer (Please click here to Add Answer)



Catch Blocks

- Catch handler must be preceded by a try block or an other catch handler
- Catch handlers are only executed when an exception has occurred
- Catch handlers are differentiated on the basis of argument type
- The catch blocks are tried in order they are written
- They can be seen as switch statement that do not need break keyword



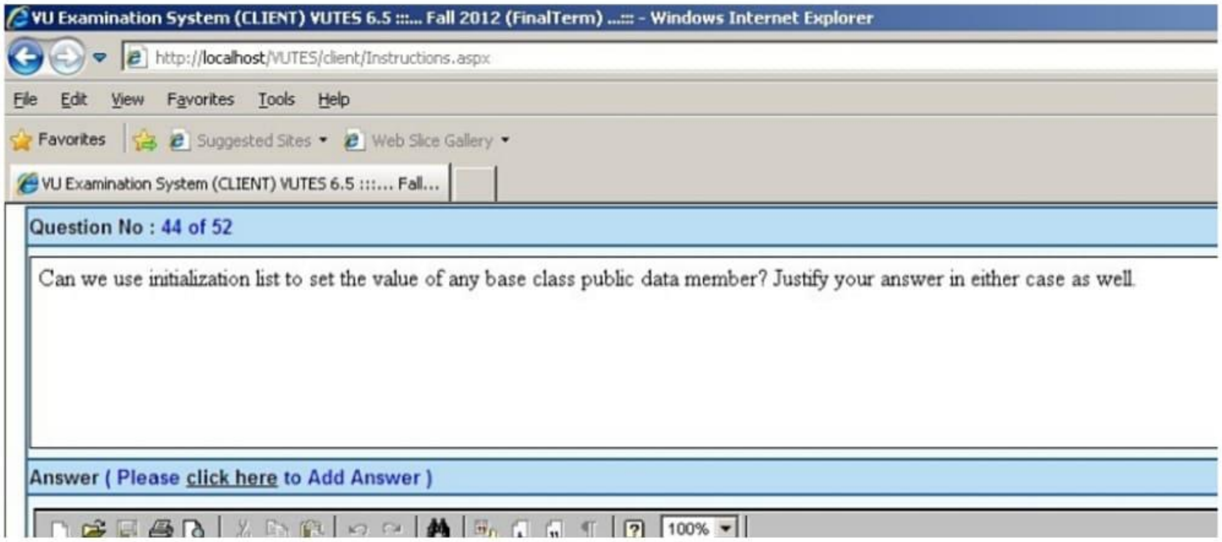
36.6.Non-type Parameters

The parameters given in template definition other than those used for mentioning templates types are called non type parameters, for example,

template <class T, class U, **int I**>

Here int I is not type parameter.

Template parameters may include non-type parameters, the non-type parameters may have default values, for example,



Initializing Members

- Derived class can only initialize members of base class using overloaded constructors
  - Derived class can not initialize the public data member of base class using member initialization list

Question No : 41 of 52

How can we set the default values for NON type parameters?

Answer (Please click here to Add Answer)

20.2.Overloading Subscript [] Operator

Subscript operator must be overloaded as member function of the class with one parameter of integer type,

```
class String{
        ...
public:
        char & operator[](int);
        ...
};

char & String::operator[]( int pos){
        assert(pos>0 && pos<=size);
        return stringPtr[pos-1];
}
int main() {
        String s1("Ping");
        cout <<str.GetString()<< endl;
        s1[2] = 'o';
        cout << str.GetString();
        return 0;
}
```

Output:

Ping
Pong

Question No : 49 of 52

The code given below has one template function as a friend of a template class,

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```
template<typename U>
void Test(U);
```

Answer (Please click here to Add Answer)

Normal | Arial | 12 | 100%

1. You have to identify any error/s in this code and describe the reason for error/s.
2. Give the correct code after removing the error/s.

```
template<typename U>void Test(U);
template< class T > class B {
        int data;
        public:
        friend void Test<>( T );
};
template<typename U>
void Test(U u){
   B < int> b1;
   b1.data = 7;
```

Muhammad Moaaz Siddiq – MCS (2nd)
mc100401285@Gmail.com
Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari

```
   }
int main(int argc, char *argv[])
{
   char i;
   Test(i);
   system("PAUSE");
   return 0;
}
```

Answer:- (correct code) Click here for detail

```
#include <cstdlib>
template<typename U> void Test(U);

template< class T > class B {
   int data;
   public:
   template <typename U > friend void Test(U);      // this statement is missing
};

template<typename U>
void Test(U u){
   B < int> b1;
   b1.data = 7;
   }
int main(int argc, char *argv[])
{
   char i;
   Test(i);
   system("PAUSE");
   return 0;
}
```

Question No : 48 of 52

What is Graceful Termination method of Error Handling; give its example using C++ code.

Answer (Please click here to Add Answer)

Normal Arial 12 100%

Q: 46: What is graceful termination method of Error handling; give its example using C++.

Answer:- (Page 330)

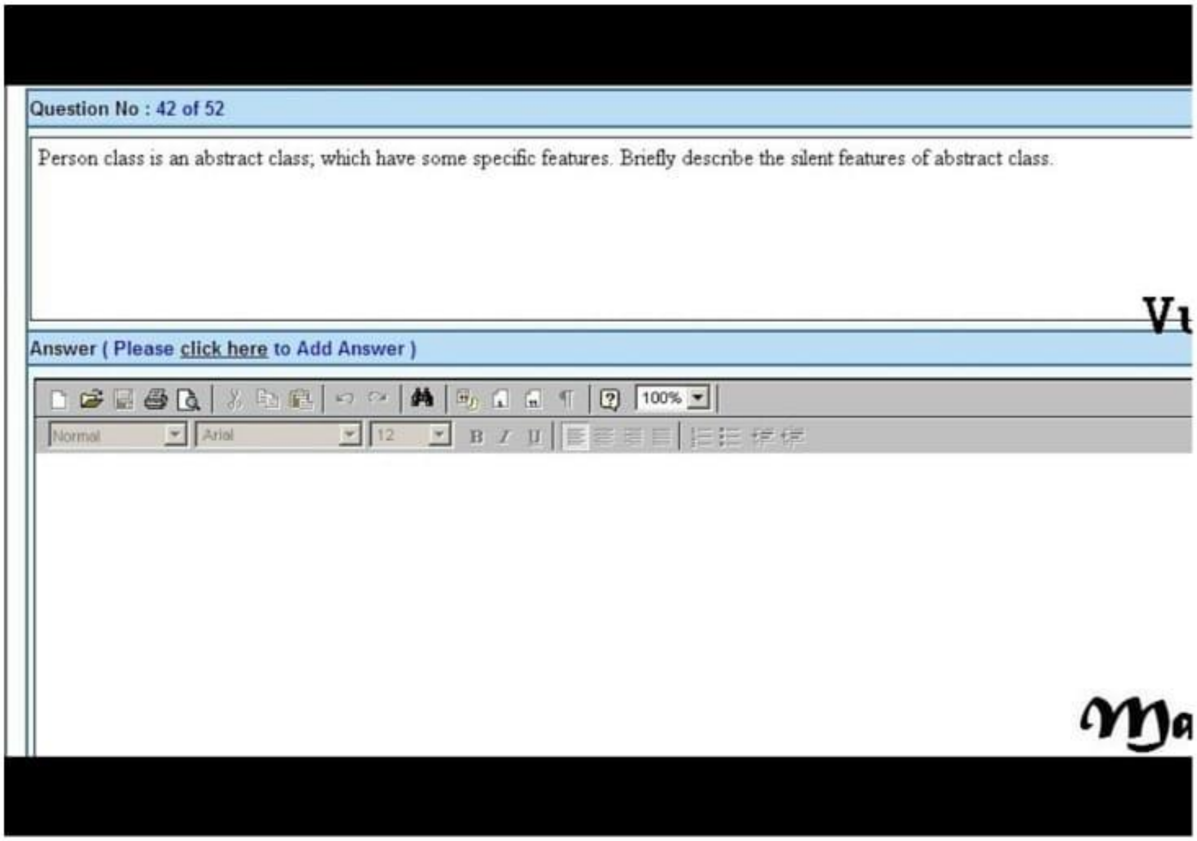
Program can be designed in such a way that instead of abnormal termination, that causes the wastage of resources, program performs clean up tasks, mean we add check for expected errors (using if conditions),

Muhammad Moaaz Siddiq – MCS (2nd)

mc100401285@Gmail.com

Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari


```
Example – Graceful Termination
int Quotient (int a, int b ) {
if(b == 0){
cout << "Denominator can't " << " be zero" << endl;
// Do local clean up
exit(1);
}
return a / b;
}
```



Describe the salint feature of abstract class (mrk3)

Answer:- (Page 230)

Abstract class's objects cannot be instantiated they are used for inheriting interface and/or implementation, so that derived classes can give implementation of these concepts.

In C++, we can make a class abstract by making its function(s) pure virtual. Conversely, a class with no pure virtual function is a concrete class

Give the C++ code of template function to print the values of any type of array I int.this function will take 2 parameters one will be pointer and other will be will be size of array (mrk3)

Answer:- (Page 257)

```
template< typename T >
void printArray( T* array, int size )
{
for ( int i = 0; i < size; i++ )
cout << array[ i ] << ", "; // here data type of array is T
}
```

[redacted]

Question No : 44 of 52

What do you mean by Stack unwinding?

Answer (Please click here to Add Answer)

100%

Normal | Arial | 12 | B I U

[redacted]

Question No: 32 (Marks: 3)
What do you mean by Stack unwinding?
Answer:- (Page 336)
The flow control (the order in which code statements and function calls are made) as a result of throw

Muhammad Moaaz Siddiq – MCS (2nd)
mc100401285@Gmail.com
Campus:- Institute of E-Learning & Modern Studies (IEMS) Samundari

statement is referred as "stack unwinding"

Question No: 33 (Marks: 3)